```
clc;
clear;
close all;
```

1. BASELINE PARAMETERS

```
serviceAreaKm2 = 5;
radiiKm = [1.0 0.5];
usersPerCell = 30;
```

2. CELL AREA AND CAPACITY CALCULATION

```
cellAreaKm2 = (3*sqrt(3)/2)*radiiKm.^2;

cellCount = ceil(serviceAreaKm2./cellAreaKm2);

estimatedUsers = cellCount*usersPerCell;
```

3. DISPLAY CAPACITY TABLE

```
Radius_km = radiiKm(:);
CellArea_km2 = cellAreaKm2(:);
EstimatedCells = cellCount(:);
UsersPerCell = usersPerCell*ones(length(radiiKm),1);
IllustrativeUsers = estimatedUsers(:);

capacityTable = table(Radius_km,CellArea_km2, ...
    EstimatedCells,UsersPerCell,IllustrativeUsers);

fprintf('\n========================================\n');
fprintf(' CELL COVERAGE AND CAPACITY RESULTS\n');
fprintf('========================================\n');

disp(capacityTable);
```

```
========================================
 CELL COVERAGE AND CAPACITY RESULTS
========================================
    Radius_km    CellArea_km2    EstimatedCells    UsersPerCell
IllustrativeUsers
    _________    ____________    ______________    ____________
_________________

         1          2.5981             2                30
60
       0.5         0.64952             8                30
240
```

4. SEVEN-CELL HEXAGONAL CLUSTER

```
R = radiiKm(1);

centers = [
    0,0
    1.5*R,sqrt(3)*R/2
    1.5*R,-sqrt(3)*R/2
    0,sqrt(3)*R
    0,-sqrt(3)*R
    -1.5*R,sqrt(3)*R/2
    -1.5*R,-sqrt(3)*R/2
];

theta = (0:6)*pi/3;

figure;
hold on;
axis equal;
grid on;

for k = 1:size(centers,1)

    hexX = centers(k,1)+R*cos(theta);
    hexY = centers(k,2)+R*sin(theta);

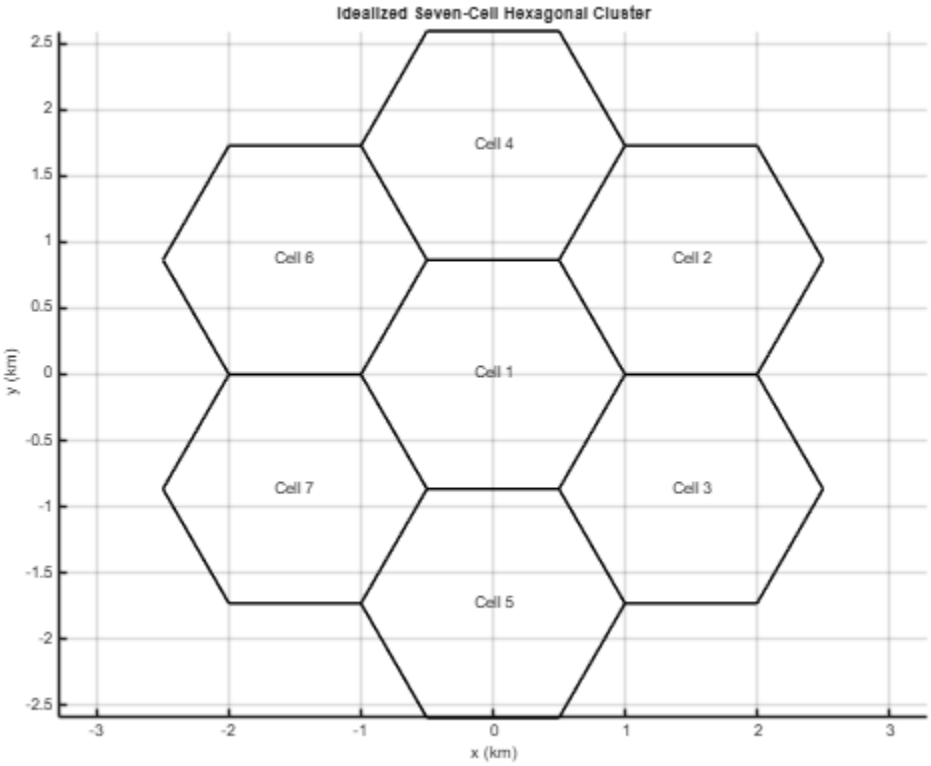
    plot(hexX,hexY,'k-','LineWidth',1.2);

    text(centers(k,1),centers(k,2), ...
        sprintf('Cell %d',k), ...
        'HorizontalAlignment','center');

end

xlabel('x (km)');
ylabel('y (km)');
title('Idealized Seven-Cell Hexagonal Cluster');

hold off;
```



5. MOBILE MOVEMENT AND HANDOFF

```
mobileX = linspace(-0.2*R,1.7*R,300);
mobileY = zeros(size(mobileX));

d1 = hypot(mobileX,mobileY);
d2 = hypot(mobileX-1.5*R,mobileY);

servingCell = ones(size(mobileX));
servingCell(d2<d1) = 2;

handoffIndex = find(diff(servingCell)~=0,1,'first');

if ~isempty(handoffIndex)
    handoffIndex = handoffIndex+1;
    handoffX = mobileX(handoffIndex);
else
    handoffX = NaN;
end
```

6. DISTANCE TO CELL CENTERS

```
figure;
hold on;
grid on;
```

```
plot(mobileX,d1,'b','LineWidth',1.5);
plot(mobileX,d2,'r','LineWidth',1.5);

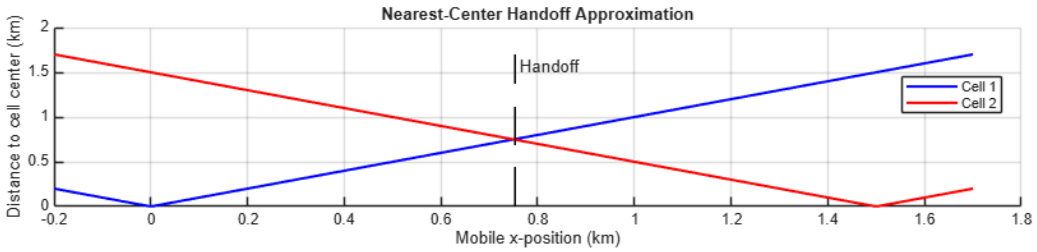
if ~isnan(handoffX)
    plot([handoffX handoffX], ...
        [0 max([d1 d2])],'k--','LineWidth',1.2);

    text(handoffX,max([d1 d2])*0.85, ...
        ' Handoff','VerticalAlignment','bottom');
end

xlabel('Mobile x-position (km)');
ylabel('Distance to cell center (km)');

legend('Cell 1','Cell 2','Location','best');
title('Nearest-Center Handoff Approximation');

hold off;
```



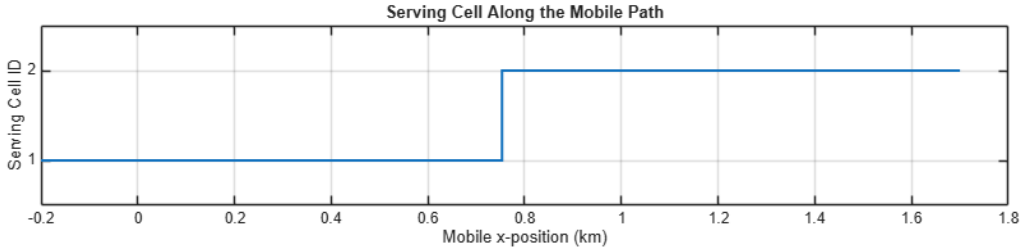
7. SERVING CELL ALONG MOBILE PATH

```
figure;

stairs(mobileX,servingCell,'LineWidth',1.5);

grid on;
xlabel('Mobile x-position (km)');
ylabel('Serving Cell ID');

title('Serving Cell Along the Mobile Path');
ylim([0.5 2.5]);
yticks([1 2]);
```



8. FINAL RESULTS

```
fprintf('\n=======================================\n');
fprintf(' HANDOFF RESULTS\n');
fprintf('=======================================\n');

if ~isnan(handoffX)
    fprintf('Geometric handoff position = %.3f km\n',handoffX);
else
    fprintf('No handoff detected.\n');
end

fprintf('\nCell radius 1 = %.2f km\n',radiiKm(1));
fprintf('Cell radius 2 = %.2f km\n',radiiKm(2));

fprintf('\nEstimated capacity for radius 1 = %d users\n', ...
    estimatedUsers(1));

fprintf('Estimated capacity for radius 2 = %d users\n', ...
    estimatedUsers(2));

fprintf('\nAll calculations and graphs completed.\n');
```

```
=======================================
 HANDOFF RESULTS
=======================================
Geometric handoff position = 0.753 km

Cell radius 1 = 1.00 km
Cell radius 2 = 0.50 km

Estimated capacity for radius 1 = 60 users
Estimated capacity for radius 2 = 240 users

All calculations and graphs completed.
```

Published with MATLAB® R2026b